Software Requirements Specification (SRS)

FanSi — AI-Powered Fashion Styling and Recommendation Website

Team Name	Hew Kao mak
Project Title	FanSi — AI-Powered Fashion Styling and Recommendation Website
SDLC Chosen & Reason	Agile Scrum — Given that the system utilizes AI to retrieve fashion trends, it is highly suitable for short iterative development cycles to gather continuous user feedback and adapt features accordingly.
Date / Version	v1.0 (25 Sep 2026)
GitHub Repo	https://github.com/pun-code/hew-kaow-makk

1. Introduction

1.1 Purpose and scope

This fashion recommendation website is designed to solve the common problem of selecting clothes and color matching. The system analyzes clothing colors and styles to create matching outfits, utilizing AI to retrieve data from various sources and summarize it into practical recommendations. It provides convenience by attaching direct purchasing links from official stores. Additionally, the system allows users to bookmark their favorite styles and share their own outfit photos to inspire the community.

1.2 Users and stakeholders

Name / Role	Description
Primary user (General users)	Seeks recommendations for color matching, styling ideas, and the convenience of clicking to purchase authentic products instantly.
Secondary stakeholder (Fashion stores)	Owners of official products linked by the system (the platform drives traffic and potential sales to their stores).
AI Provider / API	The external data source or API utilized by the system to search and analyze fashion styles.

2. Overall description

2.1 Product context — what it does / does not

FanSi is a fashion recommendation website that analyzes styles and colors alongside AI to find trends and provide official store links. Everything below defines the entire scope of the MVP for this release.

IN SCOPE	OUT OF SCOPE
• **FR-1:** Utilize AI to analyze clothing colors for matching and retrieve style data from various sources for recommendations. • **FR-2:** Display purchasing links redirecting to official stores. • **FR-3:** Bookmark system to save favorite styles for later viewing. • **FR-4:** Style sharing system allowing users to upload and post their outfits. • **FR-5:** User profile and data saving system.	• In-app shopping cart or payment gateway (the system strictly redirects to official websites). • Private chat or direct messaging systems between users. • User-to-User marketplace for selling or trading clothes.

2.2 Assumptions and constraints

- **Assumption:** Every user has a device with internet access and a functional web browser.
- **Constraint:** AI data retrieval and official store links rely on external APIs, which may be subject to daily rate limits or external downtime.
- **Constraint:** The system must implement basic data storage for user profiles, bookmarked styles, and shared posts.

3. Functional requirements

FR-1 — AI-powered style analysis and recommendation

User story: As a user, I want AI to analyze clothing colors and styles from various sources to provide recommendations, so that I get matching outfit ideas without having to think it up myself.

Pain this traces to: Users struggle to match clothing colors properly and waste time searching for ideas across multiple websites.

Acceptance Criteria:

- Given the user inputs a keyword or selects a clothing color, when submitted, the system displays style recommendations analyzed by the AI.
- Given the system displays results, when viewed, there must be clear images or descriptions of matching styles.

FR-2 — Display official store purchase links

User story: As a user, I want to see links to authentic brand products matching the recommended styles, so that I can conveniently and safely make a purchase.

Pain this traces to: Users see attractive clothes in inspiration images but do not know where to buy the authentic items.

Acceptance Criteria:

- Given a recommended clothing item is displayed, when viewed, a purchasing button or link must be attached.
- Given the user clicks the link, when executed, it must redirect to the correct official store website.

FR-3 — Bookmark favorite styles

User story: As a user, I want to bookmark styles I like, so that I can easily return to them as a reference when getting dressed later.

Pain this traces to: Users scroll past liked styles and forget them or cannot find them when they actually need to dress up.

Acceptance Criteria:

- Given the user clicks the Bookmark button on a style, when executed, that style must be added to their saved items list.
- Given the user clicks to unbookmark a style, when executed, it must disappear from the saved items list.

FR-4 — Share personal styles to the system

User story: As a user, I want to upload and share my outfit photos to the system, so that I can share ideas with other users.

Pain this traces to: Fashionable users lack a dedicated space to share their unique styles with individuals who have similar interests.

Acceptance Criteria:

- Given the user uploads an image and fills in the required details, when submitted, the system must display the post on the general feed.
- Given a post is shared successfully, when viewed on the feed, it must display the user's name or profile information.

FR-5 — User data saving system

User story: As a user, I want the system to save my profile and usage history, so that my bookmarks and shared styles are not lost when I close the web page.

Pain this traces to: Data loss when switching devices or closing the browser forces the user to search for and save styles all over again.

Acceptance Criteria:

- Given the user logs into the system, when authenticated, their previously bookmarked data must display correctly.
- Given the user closes the browser and reopens it, when returning to the site, their profile and saved styles must remain intact.

4. Non-functional requirements

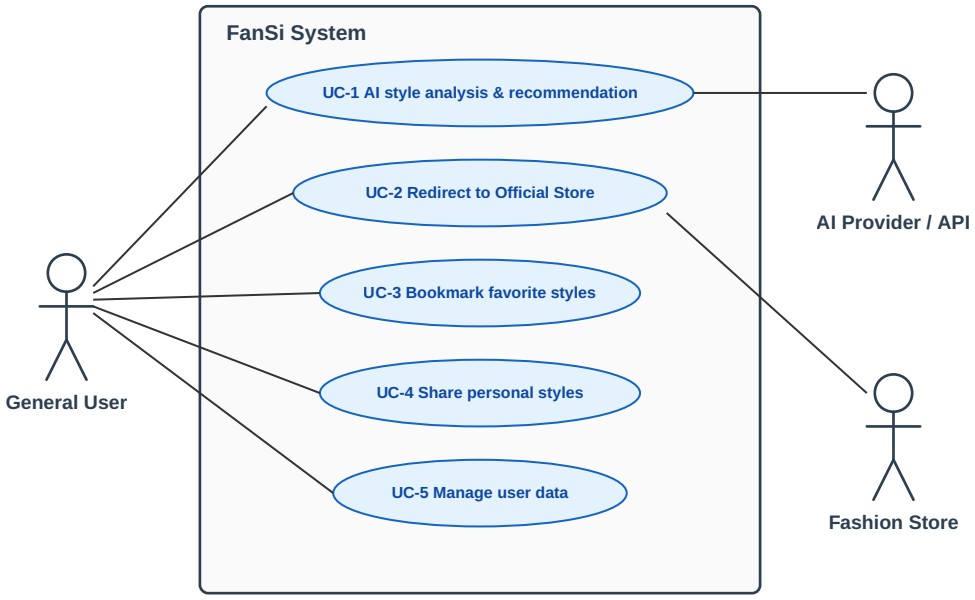
ID	Statement (Testable)	How We Will Check
NFR-1 (Performance)	AI style analysis and recommendations (FR-1) shall load within 5 seconds after the user clicks search (on a 10 Mbps internet connection or higher).	Stopwatch from clicking search to full display, 3 trials.
NFR-2 (Usability)	A first-time user shall successfully find and click the Bookmark button (FR-3) in under 15 seconds without assistance.	Timed trial with 3 users who have never seen the website before.
NFR-3 (Reliability)	At least 95% of official store links (FR-2) shall be functional, active URLs (no broken links).	Running an automated script to check all system links once a month.
NFR-4 (Security / Privacy)	A user's private bookmarked styles (FR-3) shall only be visible and accessible to the account owner.	Simulate logging in with Account A and attempting to access Account B's bookmark URL; the system must reject it (Access Denied).
NFR-5 (Compatibility)	The website shall display and execute core features (FR-1 to FR-5) perfectly on mobile Chrome and Safari browsers.	Physical device testing on iOS and Android smartphones.

5. Use cases

5.1 Use-case list

ID	Related FR & Actor	Brief Success Story
UC-1	FR-1 Actor: User	User selects the color "Brown"; the AI system processes and displays "Earth Tone" styles with images.
UC-2	FR-2 Actor: User	User clicks "Buy Authentic" on a jacket image; the system redirects to the product page on the brand's website.
UC-3	FR-3 Actor: User	User clicks the Bookmark button on a vintage style; it appears in the user's "Saved Items" page.
UC-4	FR-4 Actor: User	User uploads a photo wearing a dress, fills in details, and clicks share; the post is displayed on the general Feed.
UC-5	FR-5 Actor: User	User logs into the system; profile data and bookmark history load and display correctly.

5.2 Use-case diagram



6. Out of scope

- In-app shopping cart or payment gateway (the system strictly redirects to official websites).
- Private chat or direct messaging systems between users.
- User-to-User marketplace for selling or trading clothes.

7. Traceability matrix (Golden Thread)

Requirement (FR)	Problem	Design	Development	Test
FR-1 AI-powered style analysis	Users struggle to match colors and waste time searching for ideas.	Week 5: AI API integration and result view design.	Week 6-7: Code the AI Recommendation logic.	Week 7: Input color "Red" -> Expect matching style results.

FR-2 Official store purchase links	Users do not know where to buy the authentic items.	Week 5: Official link module and button design.	Week 6-7: Connect UI links to clothing images.	Week 7: Click link -> Redirects to official brand website.
FR-3 Bookmark favorite styles	Users forget liked styles or cannot find them later.	Week 5: Database schema design for saved items.	Week 6-7: Code Save/Unsave logic.	Week 7: Click Bookmark -> Item appears in the user's saved page.
FR-4 Share personal styles	Users lack a dedicated space to share their outfit ideas.	Week 5: Feed layout and upload system design.	Week 6-7: Code post management and image uploading.	Week 7: Upload a post successfully -> Post appears on the general feed.
FR-5 User data saving system	Data is lost when closing the browser.	Week 5: Authentication system design.	Week 6-7: Code Login and data persistence logic.	Week 7: Close browser and log back in -> Bookmark data remains intact.

8. AI usage log

Date	Tool	What we asked	What AI produced	What we changed / verified	What we learned
25 Sep 2026	Gemini	Help structure an SRS for a fashion recommendation website (FanSi) analyzing styles with AI and including share/save systems.	Full drafts for FR 1-5, NFRs, and Use cases.	Merged AI features into FR-1 and replaced FR-5 with a user data saving system to support the sharing/bookmarking logic.	AI is great at drafting user stories and Acceptance Criteria, but we must strictly control the project scope to ensure requirements match our actual constraints.
25 Sep 2026	Gemini	Verify and complete the Self-Check checklist.	Ensured all items are valid.	Verified all items against the SRS content to achieve 100% compliance.	Self-checks are vital for ensuring documentation meets professional academic standards.

9. Self-check

☑ Every FR is a user story with a clear benefit.

☑ Every FR has testable Acceptance Criteria.

☑ No FR is vague (avoiding undefined terms like "fast" or "easy").

☑ NFRs exist, including at least one covering security / privacy.

☑ Every FR traces back to a stated problem.

☑ Scope and out-of-scope are clearly stated.

☑ Use-case list matches the FR list.

☑ AI log is honest and recorded properly.